

Ayush Shinde

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SUMMARY

AI/ML Engineer with 3 years of experience across GenAI and classical ML in financial services and manufacturing. Hands-on with RAG pipelines (Python, LangChain, AWS Bedrock, OpenSearch), gradient-boosted and deep learning models (XGBoost, LightGBM, LSTM), and large-scale PySpark feature engineering on AWS. Experienced in MLOps practices spanning Airflow orchestration, MLflow tracking, drift monitoring, containerized deployment, and CI/CD.

EXPERIENCE

JPMorgan Chase & Co., TX | AI/ML Engineer | Feb 2026 – Current

- Contribute to a retrieval-augmented generation assistant over internal policy and operations documentation using Python, LangChain, and AWS Bedrock with OpenSearch vector indexing, owning retrieval evaluation and chunking strategy adopted for the production index.
- Develop and tune gradient-boosted classifiers (XGBoost, LightGBM) on 2M+ historical transaction records for anomaly and risk-flag detection, improving PR-AUC by 9% over the incumbent baseline and documenting assumptions for Model Risk review.
- Build PySpark feature pipelines on AWS EMR orchestrated through Airflow and instrument MLflow experiment tracking with post-deployment drift checks, cutting training data prep from 6 hours to 45 minutes.

Siemens, India | ML Engineer | Aug 2022 - Jul 2024

- Developed predictive maintenance models using LSTM and XGBoost, achieving 94% fault detection accuracy across 5 manufacturing plants and reducing unexpected equipment downtime by 27%.
- Engineered PySpark pipelines processing 500GB+ of IoT telemetry from 10,000+ edge devices, reducing data preparation time by 60% for downstream ML workloads.
- Built containerized inference APIs with Docker and Azure AKS, enabling real-time anomaly detection for 2,000+ production assets while reducing inference response time by 40%.
- Migrated on-premise ML infrastructure to AWS (S3, EMR, SageMaker), lowering annual infrastructure costs and improving deployment scalability.
- Built Tableau dashboards adopted by 50+ plant managers to support maintenance planning and cost-reduction decisions.
- Implemented CI/CD pipelines using GitHub Actions, reducing deployment errors by 50% and shortening model release cycles by 40%.

PROJECTS

Predictive Failure & Root Cause Agent | (PyTorch, XGBoost, LangGraph, FAISS, MLflow, FastAPI)

- Built a multi-source AI reliability system (telemetry, logs, maintenance docs): XGBoost/PyTorch models flag 85% of 462 held-out failures a median 20 h ahead (0.18 false alerts per machine-month), feeding a LangGraph root-cause agent that cites its evidence.
- Designed an evaluation harness (F1/AUROC, tool selection, retrieval, groundedness, latency, cost); on a local Qwen2.5-7B the LangGraph agent beat a ReAct baseline: root-cause accuracy 65% to 80%, unsupported responses 15.6% to 3.1% (32 questions).

RAG Support Insights & LLM Evaluation | (Python, LangChain, FAISS, Ollama, FastAPI, Streamlit)

- Built a RAG triage system over 48,178 support tickets (FAISS, local LLM via Ollama) that classifies each issue and recommends a cited next-best-action, escalating to a human through a two-stage abstention gate when evidence is weak.
- Evaluated on 200 held-out queries never indexed: the gate answers 87% of queries at 8.6% error (18.5% if it answered everything) and escalates 98% of out-of-scope messages; documented typo-robustness limits and failure cases.

SKILLS

Programming & Data: Python, SQL, PySpark, Bash, Git, Data Structures & Algorithms

Machine Learning: Supervised & Unsupervised Learning, XGBoost, LightGBM, scikit-learn, Regularization, Feature Engineering, Time-Series Forecasting, Model Evaluation (F1, ROC-AUC, PR-AUC, Precision/Recall)

Deep Learning: PyTorch, TensorFlow/Keras, Neural Networks (CNNs, LSTMs, Transformers), Hugging Face Transformers

Generative AI & LLMs: RAG Architecture, LangChain, LangGraph, Prompt Engineering, Fine-Tuning (LoRA, QLoRA), Embeddings & Semantic Search, Vector Databases (OpenSearch, FAISS, Pinecone), LLM Evaluation (groundedness, retrieval precision)

Data Engineering: Apache Airflow, Apache Spark, Kafka, AWS EMR, Batch & Streaming Pipelines, ETL

MLOps & Deployment: MLflow, Docker, Kubernetes, FastAPI, CI/CD (GitHub Actions, Jenkins), Model Monitoring & Drift Detection, Weights & Biases

Cloud: AWS (Bedrock, SageMaker, S3, EMR, Lambda), Azure ML

Math & Statistics: Linear Algebra, Probability & Statistics, Calculus, Optimization

EDUCATION

Master of Science in Data Science | University of Texas at Arlington, Arlington, TX | **May 2026**

B.E. - Electronics & Telecommunication Engineering | **AI & ML Honors** | University of Mumbai | **Aug 2024**

CERTIFICATIONS

Certified Associate Machine Learning Engineer – AWS

Databricks Fundamentals – Databricks

Advanced Google Data Analytics – Google